

COUNTING ARRANGEMENTS: CASES AND RESTRICTIONS

20/09/25

Work on file paper

1. For lunch at a particular pub you can choose between 8 different sandwiches, or 6 hot mains. If you get a sandwich then it can be accompanied by either chips, soup, crisps, or a salad. If you get a hot main then it can be accompanied by either chips, boiled potatoes, or a salad. How many different lunches are possible?
2. There are 6 balls in a bag, coloured yellow, green, brown, blue, pink and black. A ball is taken out and placed on a table. If it is yellow no further balls are removed, otherwise another ball is taken out and placed on the table next to the first ball. If the second ball is yellow, no further balls are taken, otherwise a third is taken and so on. How many possible sequences of balls are there that can end up on the table?
3. How many different sequences of the numbers $\{1, 2, 3\}$ of length 10 do not contain any of the subsequences 12, 23, or 31? (For example, 3222132111 is such a sequence).
4.
 - a. How many four-digit even numbers can be made from the numbers 0,1,2,3,4,5,6 if no digit may be repeated.
 - b. How many four-digit multiples of five can be made from the numbers 0,1,2,3,4,5,6 if no digit may be repeated.
 - c. How many four-digit numbers consisting of the digits 0,1,2,3,4,5,6 with no digit repeated are multiples of 2 or 5?
5. Six seats on a train are arranged with three seats facing the other three. In how many different ways can a family of 6 be seated if two (specific) children insist on facing forwards, and one (specific) child insists on facing backwards?
6. Eight pupils and two teachers go out to the theatre on a boarders outing. Seats have been booked in one line of 6, and one line of 4 in a different part of the theatre. How many ways of assigning the seats are there if:
 - a. the two teachers want to sit together;
 - b. one teacher must be in each line?
7. A 7 digit phone number $d_1d_2d_3d_4d_5d_6d_7$ is called *memorable* if $d_1d_2d_3$ is exactly the same sequence as $d_4d_5d_6$ or $d_5d_6d_7$ (possibly both). (For example, 4357435, 3883881, and 5555555 are memorable). Assuming each d_i can be any decimal digit (in particular, d_1 could be 0), how many memorable telephone numbers are there?
8. When I take my 12 Lower Shell tutees out to the cinema they sit in a line of 12 adjacent seats. How many ways are there of doing this if:
 - a. the two naughtiest boys mustn't sit together;
 - b. no two of the three naughtiest boys may sit together?
9. A bag contains 9 red balls, 6 blue balls, and 4 yellow balls. In how many different ways can six of these balls be placed in a line (if the only thing you can distinguish between the balls is their colour).
10. In how many arrangements of the letters ISOGRAM are:
 - a. none of the vowels together;
 - b. [Possibly hard] no three (or more) consonants together?
11. Car registration plates in Allentown consist of three letters followed by three numbers (for example, ABC 123). The letter O and the digit 0 are allowed to appear individually, but plates containing both the letter O and the number 0 are not permitted. How many allowable plates are there?

Harder problems

12. I have written five different difficult combinatorics questions to which I don't know the answers. In how many ways can I assign these questions to the top three pupils in the Remove in such a way that each of these people has at least one question to solve?
13. Generalise the result of the previous question.
14. [Hard] My Sixth Form class of twelve sits a test. I want them to mark each other's work, so I collect in the papers and hand them back to different people for marking. In how many ways can I do this, given that no pupil is to mark his own test?
15. Generalise the result of the previous question.